

JUNHAO WANG

Buffalo, NY 14260, United States

☎ (716)-495-9430 ✉ jwang436@buffalo.edu

Education

University at Buffalo, SUNY

Geological Sciences, Master of Science

08/2025 – Present

GPA: 4.0/4.0

Nanjing University of Information Science & Technology (NUIST) 09/2021 – 06/2025

Physical Geography and Resource Environment, Bachelor of Science

GPA: 87/100

Research Experience

Actionable & Computational Terrestrial Hydrology Lab

08/2025 – 2025

Supervisor: Dr. Yifan Cheng

Buffalo, United States

- Investigate historical and projected changes in Alaska's water balance and runoff-to-precipitation ratio (R/P) using 4-km coupled land-atmosphere simulation results.
- Develop a Budyko-based attribution framework to quantify the effects of precipitation, potential evapotranspiration, terrestrial water storage, and basin characteristics on R/P changes.

Climate impacts on sea ice season and ecosystem response

07/2024 – 10/2024

Supervisor: Prof. Paul Myers (Canada MITACS research intern)

Edmonton, Canada

- Analyzed changes in sea ice concentration and thickness and their spatial distribution in Baffin Bay and Hudson Bay.
- Evaluated ice-free season variability using breakup and freeze-up diagnostics.
- Examined correlations between sea ice variability and biogeochemical patterns.

Land use changes and evaluation of habitat quality

03/2023 – 04/2024

Supervisor: Prof. Chengyi Zhao; Dr. Yaojie Liu

Nanjing, China

- Conducted habitat quality assessment in China's Coastal Zone by coupling PLUS and InVEST models to predict land-use change from 2020 to 2060 under three SSP scenarios.
- Quantified driving forces of habitat degradation and contributions from land-cover transitions.

Anthropogenic climate change impact on runoff

09/2022 – 02/2024

Supervisor: Prof. Buda Su

Nanjing, China

- Investigated discharge changes in the upper Yangtze River basin during 1900–2099 under multiple forcing scenarios.
- Compared anthropogenic climate impacts with natural variability influences on discharge patterns.

Awards

Outstanding Graduate Award 2025

Canada MITACS Globalink Research Internship Award (10,000 Canadian Dollar) 2024

Geospatial Big Data and GeoAI Graduate Summer School | *Excellent Student Scholar* 2023

NUIST First Class Scholarship 2022

NUIST Merit Student 2022

Publications

- Wan, Z., Zhao, C., Zhu, J., Ma, X., Chen, J., & **Wang, J.** (2024). Assessment and Prediction of Coastal Ecological Resilience Based on the Pressure–State–Response (PSR) Model. *Land*, 13(12), 2130. <https://doi.org/10.3390/land13122130>

Research Interests

Earth System Modeling

Climate Change Impacts and Adaptation

Arctic Hydrology

Teaching Experience

ERT 101: Natural Hazards (Grader)

08/2025 – Present

Department of Earth Sciences, University at Buffalo, SUNY

Buffalo, United States

- Graded assignments and exams for an undergraduate natural hazards course.
- Held weekly office hours to support students with coursework.

Technical Skills

Coding: Python, R, Matlab, Linux shell

Tools: ArcGIS Pro, ENVI, Google Earth Engine, SPSS

Models: VIC Model, SWAT Model,

Leadership Experience

New Media Department, Changwang School of Honor, NUIST

09/2022 – 06/2023

Department Head

Nanjing, China

- Led a student media team responsible for campus digital content production and outreach.
- Coordinated event coverage, graphic design, and social media editorial for university activities.
- Managed task allocation and timelines for a multi-member student team.